

POLICYCITE-RAG

Verifiable AI for Healthcare Payment Policy

Aanya Lari | Cotiviti Intern Assessment | Topic 3



Statement-level evidence



Exact policy excerpts



Human-review routing

THE CRITICAL GAP IN PAYMENT POLICY AI

Source list ≠ Proof

A relevant document does not guarantee a supported statement. One missing qualifier can change the entire payment outcome.

Statement-Level Audit

PolicyCite-RAG verifies each claim against exact evidence, checking source authority and effective dates.

Technical Necessity

Verification must be granular to prevent claim denials or prior-authorization errors caused by AI hallucinations.



Sources: Gao et al. (2023); Chen et al. (2023). Benchmarks: ALCE, RGB.

EVOLVING RESEARCH IN AI GROUNDING

METHOD	CORE IDEA	FIT FOR POLICYCITE
MedCite	Statement-level citations & multi-pass retrieval	High: Matches need for granular evidence
Self-RAG	Adaptive retrieval and self-critique	Medium: Risk of shared model blind spots
CRAG	Correct weak retrieval via evaluation	Medium: Triggers correction when evidence is low
RARR	Research and revise unsupported text	Low: May hide original generation failures

Opportunity: A Governed Evidence Layer

Cotiviti can combine these methods with **approved sources**, **version metadata**, and **audit records**. This ensures that every AI response is grounded in a verifiable, human-governed evidence layer.

MedCite: Can Language Models Generate Verifiable Text for Medicine?

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Abstract

Existing LLM-based medical question-answering systems lack citation generation and evaluation capabilities, raising concerns about their adoption in practice. In this work, we introduce MedCite, the first end-to-end framework that facilitates the design and evaluation of citation generation with LLMs for medical tasks. Meanwhile, we introduce a novel multi-pass retrieval-citation method that generates high-quality citations. Our evaluation highlights the challenges and opportunities of citation generation for medical tasks, while identifying important design choices that have a significant impact on the final citation quality. Our proposed method achieves superior citation precision and recall improvements compared to strong baseline methods, and we show that evaluation results correlate well with annotation results from professional experts.

1 Introduction

Large Language Models (LLMs) have demonstrated remarkable capabilities in various natural

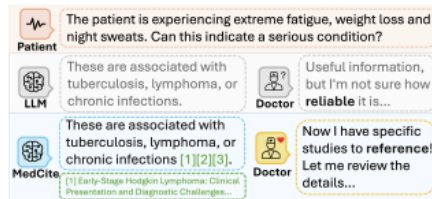


Figure 1: Medical QA system comparison. State-of-the-art systems generate answers without citations. MedCite not only generates answers but also associates each answer with citations, improving the verifiability and trustworthiness of the medical system.

or misleading, pose significant challenges to the reliability of LLM-based medical systems (Pal et al., 2023; Ahmad et al., 2023; Huang et al., 2024). To overcome the issue, researchers and practitioners have studied retrieval-augmented generation (RAG) (Xiong et al., 2024a; Yang et al., 2024), which combines LLMs with information retrieval from external trustworthy data sources (Canese and Weis, 2013). By providing the model with accurate and relevant medical knowledge, these systems

MEDCITE: A VALIDATED FOUNDATION



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Core Contributions

- **End-to-End Framework**

A comprehensive method for generating and evaluating medical citations in LLM responses.

- **Multi-Pass Retrieval**

Iterative search strategy that improves citation precision and recall over strong baselines.

- **Expert Alignment**

Evaluation results correlate strongly with annotations from professional medical experts.



Fit for PolicyCite-RAG

- **Statement-Level Evidence**

Directly addresses the need for granular verification in healthcare payment policy.

- **Metric Adaptation**

MedCite's attribution and citation metrics are adapted for the CMS policy corpus.

- **Research Inspiration**

Provides a credible, peer-reviewed foundation for the POC's technical architecture.

A TRANSPARENT TECHNICAL ARCHITECTURE

CORPUS & RETRIEVAL

CMS Policy Corpus

Three public CMS documents stored as JSONL.

Hybrid Retrieval

FAISS (Dense) + BM25 (Lexical) for high recall.

RRF Integration

Reciprocal Rank Fusion for optimized ranking.

APPLICATION LAYER

Local LLMs (Ollama)

Llama 3.2 (1B) for generation; Qwen 2.5 (3B) for audit.

FastAPI Backend

Exposes baseline and cited-answer paths.

Streamlit UI

Live dashboard for audit trail visualization.

AUDIT CONTROLS

Provenance Separation

Generation and audit evidence stored independently.

Binary Verdicts

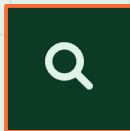
Deterministic support/no-support classification.

Review Routing

Automated flags for human-in-the-loop validation.

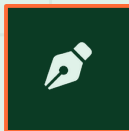
*Stack intentionally avoids complex agents to maintain a deterministic and inspectable audit trail.

SEPARATING GENERATION FROM PROVENANCE



Hybrid Retrieval

FAISS + BM25 search across CMS policy corpus.



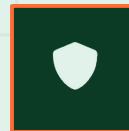
Answer Generation

Baseline answer created from initial passages.



Statement Extraction

Answer parsed into individual factual claims.



Evidence Audit

Dedicated re-retrieval for each statement.

Independent Verification: Evidence retrieved after generation is stored separately to prevent mislabeling.

Audit Provenance: The system substantiates statements without confusing audit evidence with generation context.

RIGOROUS CITATION-ASSURANCE CONTROLS



Deterministic Validation

- **ID Verification:** Cross-references returned citation IDs against the master policy corpus.
- **Verbatim Extraction:** Copies exact excerpts directly from source JSONL to prevent paraphrasing.
- **Metadata Mapping:** Converts internal indices to user-facing display pages and effective dates.



Model-Based Attribution

- **Qualifier Audit:** Ensures every date, quantity, and modality is explicitly supported by the text.
- **Scope Check:** Validates that exclusions and conditions in the answer match the policy evidence.
- **Binary Verdicts:** Forces a strict support/no-support decision to avoid ambiguous grounding.



Human-Review Routing

Any statement flagged as **NOT_SUPPORTED**—due to missing, partial, or conflicting evidence—is automatically routed to a human reviewer. This prevents the system from inventing metadata or presenting hallucinations as facts.

PERFORMANCE AND LATENCY EVALUATION

100%

AUDITED CITATION
PRECISION & COVERAGE

-25%

REDUCTION IN
REDUNDANT CITATIONS

0.15s

MEDIAN LATENCY
OVERHEAD

Artifact: [evaluation/results/20260702T041034Z.json](#)

Scope: 10 questions | 3 documents | Author signoff complete.

PolicyCite demonstrates technical feasibility for a compact, inspectable audit trace with minimal performance impact.

STRATEGIC PILOT RECOMMENDATION



1. Bounded Domain Start

Begin with one versioned policy domain restricted to approved, effective-dated sources.

2. Reviewer-in-the-Loop

Require explicit evidence for every claim and route all flags to human review.

3. Versioned Audit Records

Preserve statement, evidence, corpus versions, and reviewer decisions in a permanent log.

4. Success Metrics

Measure qualifier preservation, expert agreement, and reviewer time saved.

Core Strategy

Build the evidence layer before expanding the chatbot.

Expansion should be contingent on high citation precision and measurable workflow value.



POLICYCITE-RAG

DEMO



Thank you

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